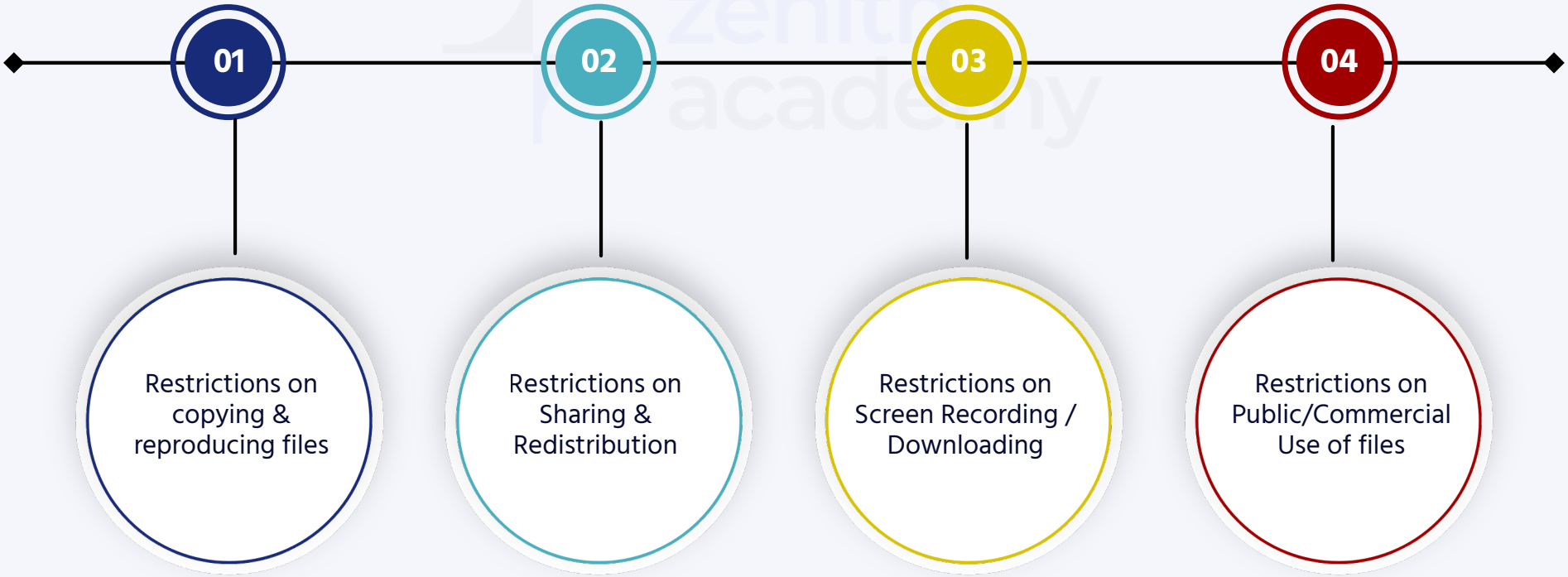
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ACUTE DIARRHOEAL DISEASES

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DR.ARDRA MERIN GEORGE

DIARRHOEA

- Passage of loose, liquid or watery stools
- Usually passed more than 3 times a day
- Recent change in the consistency and character of stools

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CLINICAL TYPES OF DIARRHOEAL DISEASE

ACUTE WATERY DIARRHOEA	ACUTE BLOODY DIARRHOEA	PERSISTENT DIARRHOEA	DIARRHOEA WITH SEVRE MALNUTRITION
• Several hours to days • Dehydration and weight loss • Vibrio cholerae • Escherichia Coli • Rota virus	• Dysentery • Damage to intestinal mucosa, sepsis and malnutrition • dehydration • Shigella	• 14 days or longer • Malnutrition and serious non – intestinal infection, dehydration • AIDS	• Severe systemic infection • Heart failure • Vitamin and mineral deficiency • Seen in marasmus and kwashiorkor

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DIARRHOEAL PREVENTION INDICATORS

Drinking water and sanitation

- improved drinking water sources
- Improved sanitation facilities

Immunization

- % of one year old child immunized against measles

Nutrition

- Under weight(moderate and severe) 0-59 months
- Stunted(moderate and severe) 0-59 months
- Exclusively breastfed 0-5 months
- Breastfed with complementary food 6-9 months
- Still breast feeding 20-23 months

Vitamin A

- Vitamin A supplementation full coverage rate 6-59 months

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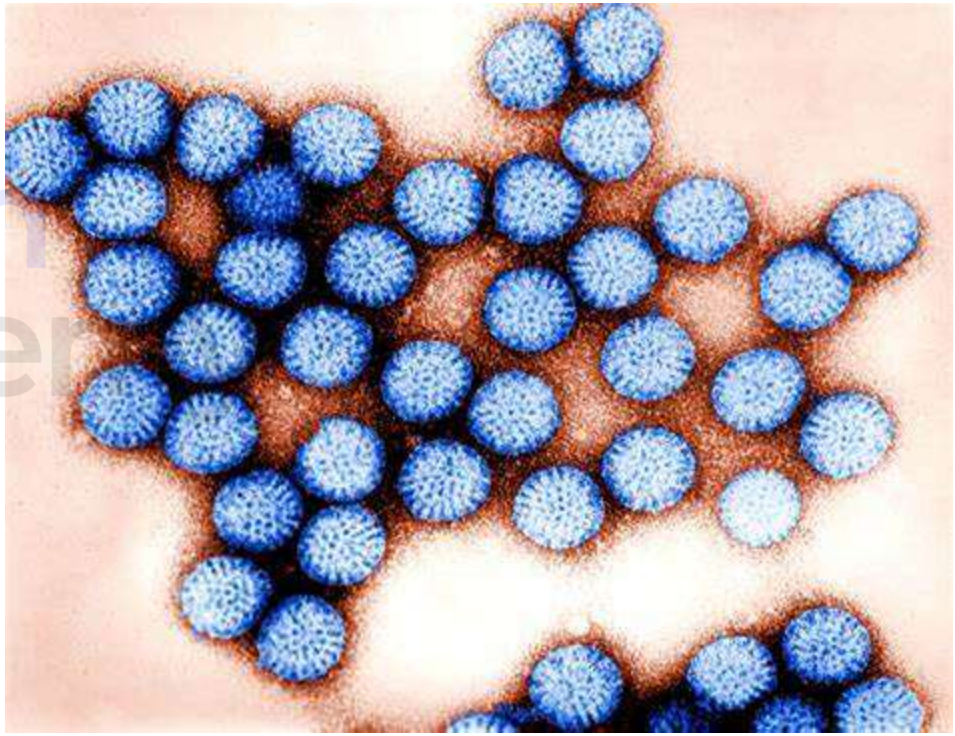
DIARHHOEA TREATMENT INDICATORS (% OF CHILDREN UNDER 5)

- ORT with continuous feeding
- ORS packet
- Recommended homemade fluids
- Increased fluids
- Continued feeding

AGENT FACTORS

ROTA VIRUS

- Discovered in 1973
- Leading cause of diarrhoea among under 5 globally
- Re infection common
- High concentrations shed on stool and vomitus
  - Faecal oral route
  - Person to person
  - Contaminated fomites



BACTERIAL CAUSES

Escherichia Coli (ETEC)

- Adults and children
- Most common cause of traveller's diarrhoea
- Doesn't invade mucosa
- Toxins released
- 2 toxins
  - Heat labile(LT)
  - Heat stable(ST)
- Spread via contaminated food and water

Vibrio cholerae

- 5-10 % of diarrhoeas
- no inflammation of bowel epithelium

- Salmonella causes inflammation of bowel epithelium

Shigella

- High mortality
- Dysentery

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OTHERS

Amoebiasis

Giardiasis

- Duodenum and jejunum

Cryptosporidium

- Parasite
- Diarrhoea in infants ,immunocompromised. domestic animals
- AIDS- persistent diarrhoea with wasting

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RESERVOIR

- HUMANS
  - E.coli
  - Shigella
  - Cholerae
  - Giardia
  - Entamoeba histolytica

- ANIMALS AND HUMANS
  - Campylobacter jejuni
  - Salmonella
  - Y. enterocolitica

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HOST FACTORS

- 6 months – 2 years
- Most common 6-11 months (declining levels of maternal immunity and weaning)
- Common in people with malnutrition
- Poverty
- Prematurity
- Reduced gastric acidity
- Immunodeficiency
- Lack of personal hygiene and domestic hygiene
- Incorrect feeding practices

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ENVIRONMENTAL FACTORS

- Temperate climates →bacterial diarrhoea → warm season
- →Viral diarrhoea →winter

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- Tropical climates →bacterial diarrhoea → warmer and rainy season
- →rotavirus →throughput year → more in cool months

MODE OF TRANSMISSION

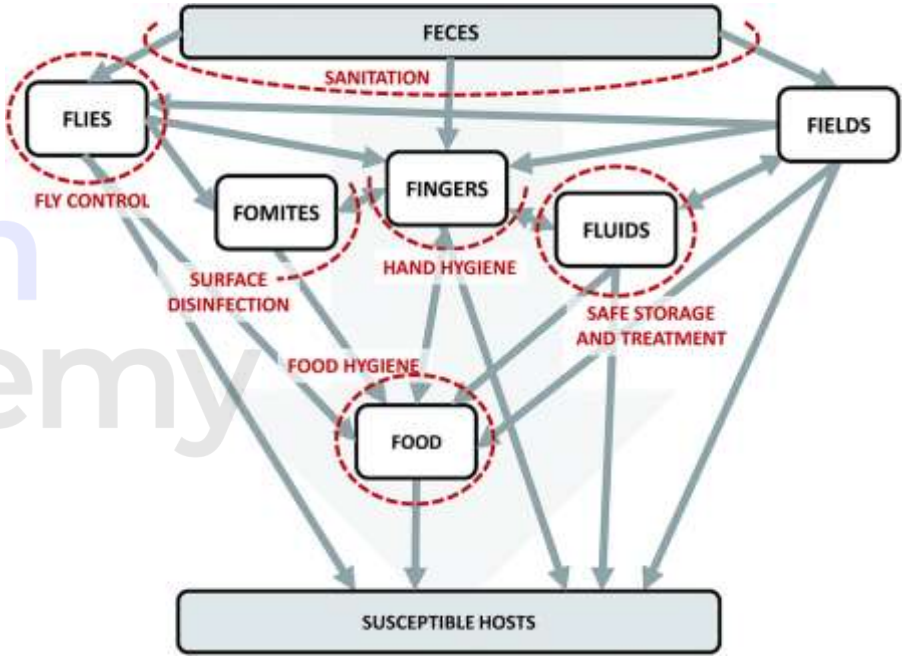
FACECAL –ORAL ROUTE

Water borne

Food borne

Direct transmission

- Fingers
- Fomites
- Dirt



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1. Diarrhoea is clinically defined as the passage of loose, liquid, or watery stools at least how many times a day?

A. Two times

B. Three times

C. Five times

D. Seven times

Answer: B

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2. Persistent diarrhoea is defined as an episode of diarrhoea that lasts for:

A. 7 days or longer

B. 10 days or longer

C. 14 days or longer

D. 21 days or longer

Answer: C

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3. Which pathogen is most commonly responsible for "traveler's diarrhea" in both adults and children?

A. *Shigella*

B. *Vibrio cholerae*

C. Enterotoxigenic *E. coli* (ETEC)

D. Rotavirus

Answer: C

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4.Which clinical type of diarrhoea is primarily characterized by the danger of intestinal mucosa damage, sepsis, and malnutrition?

A. Acute watery diarrhoea

B. Acute bloody diarrhoea (Dysentery)

C. Persistent diarrhoea

D. Diarrhoea with severe malnutrition

Answer: B

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CONTROL OF DIARRHOEAL DISEASES

Diarrhoeal Disease Control Programme -1980

SHORT TERM

- Appropriate clinical management

LONG TERM

- Better MCH care practices
- Preventive strategies
- Preventing diarrhoeal epidemics

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APPROPRIATE CLINICAL MANA

- Prevent dehydration and reduce mortality
- Glucose will enhance the intestinal absorption of salt and water
- India is the first country in the world to launch reduced osmolarity ORS June 2004

Composition of reduced osmolarity ORS

REDUCED OSMOLALITY ORS	GRAM/ LITRE
SOD.CHLORIDE	2.6
GLUCOSE, ANHYDROUS	13.5
POTASSIUM CHLORIDE	1.5
TRISODIUM CITRATE , DIHYDRATE	2.9
TOTAL WEIGHT	20.5

REDUCED OSMOLARITY ORS	Mmol/L
SODIUM	75
CHLORIDE	65
GLUCOSE , ANHYDROUS	75
POTASSIUM	20
CITRATE	10
TOTAL OSMOLARITY	245

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Clinical Assessment of Dehydration: Identifying Mild vs. Severe Cases



Mild Dehydration (Caution - Plan A/B)	Severe Dehydration (Urgent - Plan C/IV)
General Appearance Thirsty, alert, restless	**General Appearance** Drowsy, limp, cold, sweaty, comatose
Radial Pulse Normal rate and volume	**Radial Pulse** Rapid, feeble, or impalpable
Blood Pressure Normal	**Blood Pressure** < 80 mm Hg or unrecordable
Skin Elasticity (The Pinch Test) Retracts immediately	**Skin Elasticity** (The Pinch Test) Retracts very slowly (> 2 sec)
Tongue Moist	**Tongue** Very dry
Ant. Fontanelle Normal	**Ant. Fontanelle** Very sunken
Urine Flow Normal	**Urine Flow** Little or none
Percentage of Body Weight Loss 4–5% loss	**Weight Loss** 10% or more loss
Estimated Fluid Deficit 40–50 ml/kg	**Estimated Fluid Deficit** 100–110 ml/kg

NotebookLM

- If the child's weight is known the amount of ORS required for rehydration during the first 4 hours may be calculated at approximately 75ml/Kg
- The amount given depends on patients desire to drink
- More amount to heavier patients and with more severe signs
- Older children should be given as much as water they want in addition to ORS

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AFTER EACH STOOL

Children under 2 years of age → give a teaspoon every 1-2 minutes

→ 50-100 ml (a quarter to half a large cup) after each stool

2-10 years →100-200ml (half to one large cup)

Older children and adults > drink as much as they want

If child vomits → wait for 10 minutes and try again

A spoonful every 2-3 minutes

If the child is breastfed → nursing pursued during treatment with ORS

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ORS

- The solution should be made fresh every day and used within 24 hours
- One teaspoon salt and 6 teaspoon sugar dissolved in 1 litre of drinking water
- Infants usual diet should be continued during diarrhoea and increased afterwards
- Food should never be with held and should never be diluted

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PLAN A

- 4 rules of home treatment
- Give extra fluid
- Give zinc supplements
- Continue feeding
- When to return
- **No Dehydration**

PLAN B

- Determine the ORS to be given in four hours
- Show the mother how to prepare ORS
- After 4 hours – reassess
- 4 rules of home treatment
- **Some dehydration**

PLAN C

- Start IV fluids immediately
- Ringer lactate
- Refer urgently to hospital
- Reassess
- **Severe dehydration**

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Pediatric Diarrhea Management: A Guide to Plans A, B, and C

Plan A: Home Care for No Dehydration

Give extra fluids (ORS, salted soups, rice water), continue feeding, and provide Zinc supplements for 10–14 days.

Zinc Dosage (Daily for 10–14 Days)

Child's Age	Zinc Dosage
Under 6 Months	10 mg
Over 6 Months	20 mg

Plan B: Oral Rehydration for "Some" Dehydration

Give ORS sips at a health facility for 4 hours; if vomiting occurs, wait 10 minutes then continue slowly.



Plan C: Emergency Care for Severe Dehydration

Administer immediate IV fluids (Ringer's Lactate or Normal Saline) at a rate of 100ml/kg.



When to Return Immediately

Physical Danger Signs

- Many watery stools
- Fever
- Blood in the stool

Intake & Hydration Red Flags

- Marked thirst
- Poor eating/drinking
- Repeated vomiting

NotebookLM

PLAN A

Give extra Fluid	Give zinc	Continue feeding	When to return
• Breastfed frequently • ORS, home based fluids • Teach how to mix and give ORS • How much fluid to give in addition to the usual fluid intake	• 2 months – 6 months → ½ tablet daily for 14 days • > 6 months → 1 tablet daily for 14 days	• Exclusively breastfed if < 6 months	• Danger signs

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INTRAVENOUS REHYDRATION

- Ringer lactate solution (Hartmann's solution for injection)
- Diarrhoea treatment solution
- The recommended dose of IV fluid to be given is 100ml/kg
- Initial rehydration should be fast
- Reassess every 1-2 hours
- When patient can drink give ORS → 5ml/kg/hour
- Reassess every 4 hour
- Continue till all signs have disappeared

AGE	First give 30ml/kg	Then 70ml/kg
Less than 12 months	1 hour	5 hours
Older	30 minutes	2.5 hours

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MAINTENECE THERAPY

Amount of diarrhoea	Amount of oral fluid
Mild diarrhoea (not more than one stool every 2 hours or longer)	100ml/kg body weight per day until diarrhoea stops
Severe diarrhoea (more than one stool every 2 hours)	Replace stool loses volume for volume (10-15ml/kg per hour)

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APPROPRIATE FEEDING

Normal food intake promoted

No gut rest needed

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ZINC SUPPLEMENTATION

10-14 days

<6 months 10 mg daily

>6 months 20mg daily

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THEN ASK:
Does the young infant have diarrhoea?*

IF YES, ASK:	LOOK AND FEEL:
• For how long? • Is there blood in the stool?	• Look at the young infant's general condition. Is the infant: Lethargic or unconscious? Restless and irritable? • Look for sunken eyes. • Pinch the skin of the abdomen. Does it go back: Very slowly (longer than 2 seconds)? Slowly?

Classify DIARRHOEA

for DEHYDRATION	**Two of the following signs:** • Lethargic or unconscious • Sunken eyes • Skin pinch goes back very slowly.	**SEVERE DEHYDRATION**
	Two of the following signs: • Restless, irritable. • Sunken eyes. • Skin pinch goes back slowly.	**SOME DEHYDRATION**
	• Not enough signs to classify as some or severe dehydration.	**NO DEHYDRATION**
and if diarrhoea 14 days or more	• Diarrhoea lasting 14 days or more.	**SEVERE PERSISTENT DIARRHOEA**
and if blood in stool	• Blood in the stool.	**SEVERE DYSENTERY**

Give first dose of intramuscular ampicillin and gentamicin. ***If infant also has low weight or another severe classification:*** ***- Refer URGENTLY to hospital with mother giving frequent sips of ORS on the way.*** ***- Advise mother to continue breastfeeding.*** ***- Advise mother how to keep the young infant warm on the way to the hospital.*** ***OR*** If infant does not have low weight or any other severe classification: - Give fluid for severe dehydration (Plan C) and then refer to hospital after rehydration
If infant also has low weight or another severe classification: ***-Give first dose of intramuscular ampicillin and gentamicin*** ***- Refer URGENTLY to hospital with mother giving frequent sips of ORS on the way.*** ***-Advise mother to continue breastfeeding.*** ***-Advise mother how to keep the young infant warm on the way to the hospital.*** *If infant does not have low weight or another severe classification:* - Give fluids for some dehydration (Plan B). - Advise mother when to return immediately. - Follow up in 2 days
Give fluids to treat diarrhea at home (Plan A). Advise mother when to return immediately. Follow up in 5 days if not improving.
Give first dose of intramuscular ampicillin and gentamicin if the young infant has low weight, dehydration or another severe classification. **Treat to prevent low blood sugar.** **Advise how to keep infant warm on the way to the hospital.** **Refer to hospital.#**
Give first dose of intramuscular ampicillin and gentamicin if the young infant has low weight, dehydration or another severe classification. **Treat to prevent low blood sugar.** **Advise how to keep infant warm on the way to the hospital.** **Refer to hospital . #**

* ***What is diarrhoea in a young infant?***

If the stools have changed from usual pattern and are many and watery (more water than fecal matter). The normally frequent or loose stools of a breastfed baby are not diarrhoea.

*# If referral is not possible, see the section **Where Referral Is Not Possible** in the module **Treat the Young Infant and Counsel the Mother**.*

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5. In a child with severe dehydration, the skin pinch (skin elasticity) typically retracts:

A. Immediately

B. In less than 1 second

C. Slowly (less than 2 seconds)

D. Very slowly (more than 2 seconds)

Answer: D

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6. For an infant under 12 months of age receiving IV rehydration (Plan C), the first 30 ml/kg of Ringer's Lactate should be administered over:

A. 30 minutes

B. 1 hour

C. 5 hours

D. 2.5 hours

Answer: B

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7. Zinc supplementation during an episode of acute diarrhea should be continued for how many days?

A. 3–5 days

B. 5–7 days

C. 10–14 days

D. 21 days

Answer: C

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8. For maintenance therapy in a child with mild diarrhoea, the recommended amount of oral fluid is:

A. 50 ml/kg per day

B. 100 ml/kg per day

C. 150 ml/kg per day

D. 200 ml/kg per day

Answer: B

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CHEMOTHERAPY

- Only when diarrhoea has been identified as shigella, typhoid or cholera

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Symptoms	Cholera	Shigella
Diarrhoea	Acute watery diarrhoea	Acute bloody diarrhoea
Fever	No	Yes
Abdominal cramps	Yes	Yes
Vomiting	Yes	No
Rectal pain	No	Yes
stool	> 3 loose stools per day, watery like rice water	> 3 stools per day with pus or blood
Drug of choice	Doxycycline , tetracycline, TMP-SMX and erythromycin	Ciprofloxacin

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BETTER MCH PRACTICES

MATERNAL NUTRITION

- Prenatal nutrition
- Postnatal nutrition
- Breast feeding

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CHILD NUTRITION

- Promotion of breast feeding
- Appropriate weaning practices
- Supplementary feeding
- Vitamin A supplementation -19-54% reduction in mortality. It also reduces duration, severity and complications associated with diarrhoea

PREVENTIVE STRATERGIES

SANITATAION

- Improved water supply ,improved excreta disposal, improved domestic and food hygiene
- Handwashing before and after preparing food, before feeding child ,after cleaning defecated child and after disposing off a child's stool
- Clean and functioning latrine
- If no latrine then defecate at a distance of 10m away from water supply

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HEALTH EDUCATION

- Breast feeding
- Improved weaning
- Clean drinking water
- Use pf plenty of water hygiene
- Use of latrine
- Disposal of stools of young children

IMMUNIZATION

- Measles

ROTA VIRS VACCINE

Rotarix	RotaTeq
• Live attenuated • 2 doses monovalent • Oral • 2 and 4 months • First dose can be started at 6 weeks and no later than 12 weeks • Gap -4 weeks • Completed within 16 weeks ,no later than 24 weeks of age	• Live attenuated • 3 doses Pentavalent • Oral • 2,4 and 6 months • First dose between 6-12 weeks and subsequent doses at intervals of 4-10 weeks • All 3 doses administered before 32 weeks • Vaccine should not be initiated for infants aged >12 weeks

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The Integrated Global Action Plan for the prevention and Control of Pneumonia and Diarrhoea

Specific goals for 2025

1. Reduce mortality from pneumonia <5 years to fewer than 3 per 1000 LB
2. Reduce mortality diarrhoea <5 years to fewer than 1 per 1000 LB
3. Reduce incidence of severe pneumonia by 75% in < 5 years as compared to 2010 levels
4. Reduce incidence of severe diarrhoea by 75% in < 5 years as compared to 2010 levels
5. Reduce by 40% the global number of children < 5 years who are stunted as compared to 2010 levels

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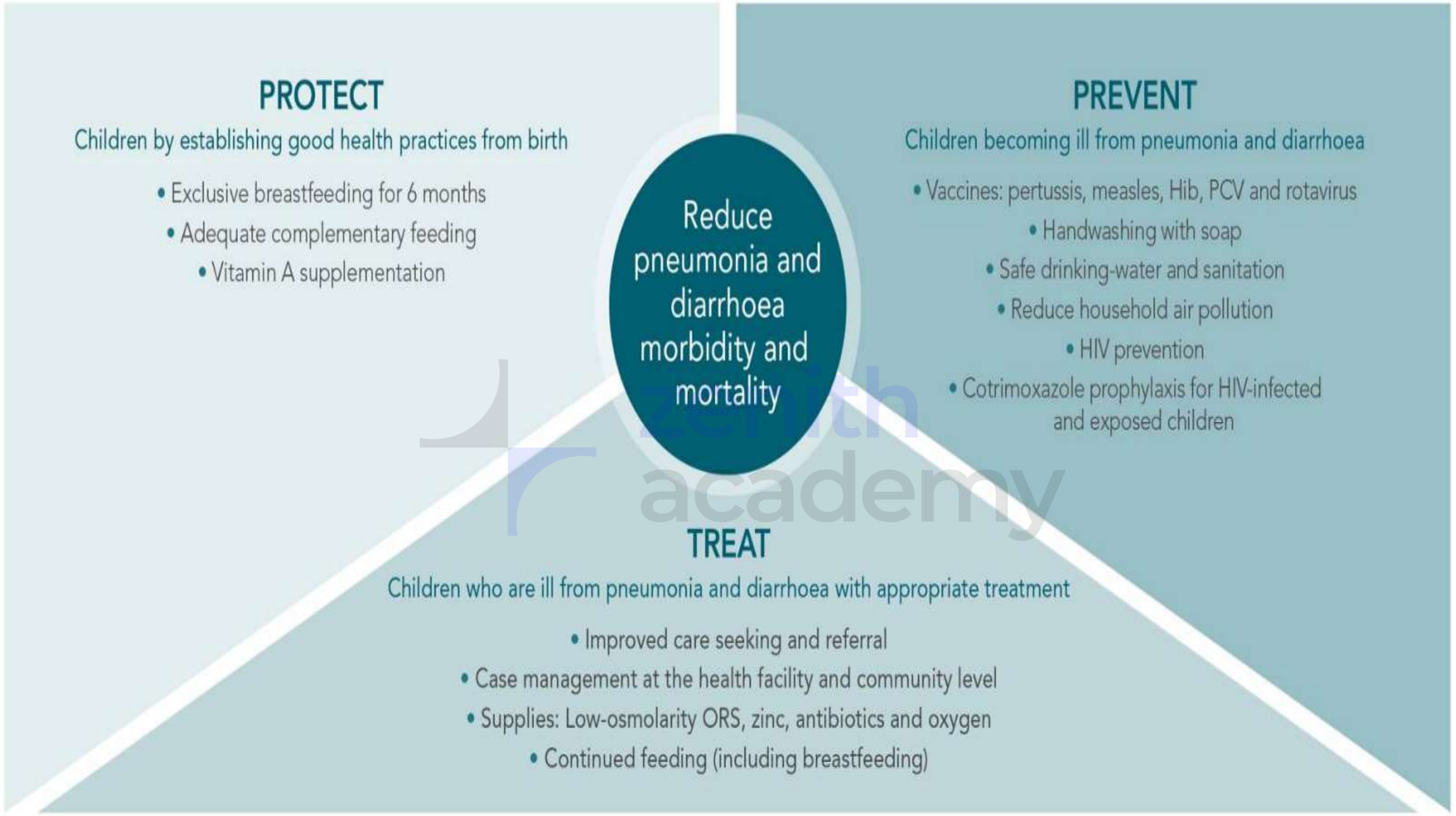
By the end of 2025

- 90% full dose coverage of each relevant vaccine (80% coverage in veery district)
- 90% access to appropriate pneumonia and diarrhoea case management (8-% coverage in each district)
- At least 50% coverage of exclusive breast feeding during first 6 months of life
- Virtual(vertical) elimination of pediatric HIV

By the end of 2030

- Universal access to basic drinking water in health care facilities and homes
- Universal access to adequate sanitation in heath car facilities by 2030 and in homes by 2040
- Universal access to handwashing facilities in health care facilities and homes
- Universal access to clean and safe energy technologies in health care facilities and homes

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PROTECT
Children by establishing good health practices from birth
• Exclusive breastfeeding for 6 months
• Adequate complementary feeding
• Vitamin A supplementation
PREVENT
Children becoming ill from pneumonia and diarrhoea
• Vaccines: pertussis, measles, Hib, PCV and rotavirus
• Handwashing with soap
• Safe drinking-water and sanitation
• Reduce household air pollution
• HIV prevention
• Cotrimoxazole prophylaxis for HIV-infected and exposed children
Reduce pneumonia and diarrhoea morbidity and mortality
TREAT
Children who are ill from pneumonia and diarrhoea with appropriate treatment
• Improved care seeking and referral
• Case management at the health facility and community level
• Supplies: Low-osmolarity ORS, zinc, antibiotics and oxygen
• Continued feeding (including breastfeeding)

1)Which among the following organism causes dysentery

A. E.Coli
B. Shigella
C. Cholera
D. Rota virus

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Answer : B shigella

2) Which among the following is not included under diarrhoea prevention indicators

A. % of population using improved drinking water
B. % of under 5 children immunized against measles
C. % of children who are moderate under weight
D. Vitamin A supplementation coverage

Answer B (under 1 year)

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3) Which among the following is not included in diarrhoea treatment indicators(% of under 5)

A. ORT with continuous feeding
B. ORS packet
C. Recommended homemade fluids
D. Increased fluids
E. Exclusively breastfed

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Answer: E

4)Which is the first country to launch ORS as per WHO and UNICEF recommendation in 2004

A. India
B. US
C. UK
D. China

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Answer: A India

5)Which among the following is false regarding the fluid replacement in diarrhoea

A. The ORS solution can be made and can be used for multiple days till it is over for the effective use

B. One teaspoon salt and 6 teaspoon sugar dissolved in 1 litre of drinking water

C. Infants usual diet should be continued during diarrhoea and increased afterwards

D. Food should never be with held and should never be diluted

Answer : A

6)What is the management for a child with severe dehydration

A. Can be sent home with oral fluids

B. Can be given ORS and reassess and send home

C. Start with replacement therapy after hospital admission

Answer: C

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7)What is the management for an infant who presented with very dry tongue, drowsiness

A. Can be sent home with oral fluids

B. Can be given ORS and reassess and send home

C. Start with replacement therapy after hospital admission

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Answer:C

8) What is the amount of fluid to be replaced for an infant for some dehydration if the weight is known

A. 100ml/kg

B. 20ml/kg

C. 75ml/kg

D. 60ml/kg

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Answer : C

9)Which among the following is the best fluid for replacement therapy for diarrhoea

A. Ringer lactate
B. Diarrhoea treatment solution
C. Normal saline
D. Dextrose

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Answer :A

10. What is the total osmolarity of the "reduced osmolarity ORS" recommended by WHO and UNICEF?

A. 311 mmol/l

B. 245 mmol/l

C. 200 mmol/l

D. 285 mmol/l

Answer: B

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11)Under the GAPPD framework, what is the global target for reducing mortality from diarrhea in children under 5 years of age by 2025?

A) Fewer than 1 per 1,000 live births

B) Fewer than 3 per 1,000 live births

C) Fewer than 5 per 1,000 live births

D) Fewer than 10 per 1,000 live births

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A) Fewer than 1 per 1,000 live births

Q12. What is the 2025 GAPPD target for reducing severe diarrhoea incidence in children under 5 years of age compared to 2010 levels?

A) Reduce by 25%

B) Reduce by 50%

C) Reduce by 75%

D) Reduce by 90%

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Correct Answer: C) Reduce by 75%

Q13. Which three core pillars form the strategic framework of GAPPD?

A) Isolate, Quarantine, Treat

B) Protect, Prevent, Treat

C) Diagnose, Vaccinate, Hospitalize

D) Monitor, Educate, Medicate

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Correct Answer: B) Protect, Prevent, Treat

Q14. Vitamin A supplementation in children aged 6 to 59 months reduces child mortality by what range?

A) 5% to 10%

B) 10% to 15%

C) 19% to 54%

D) 60% to 80%

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Correct Answer: C) 19% to 54%

Q15. What type of vaccine is Rotarix™ and how many doses are in its recommended primary schedule?

A. Killed virus vaccine, 3-dose schedule

B. Monovalent human live attenuated vaccine, 2-dose schedule

C. Pentavalent bovine reassortant vaccine, 3-dose schedule

D. Subunit protein vaccine, 1-dose schedule

Correct Answer: B) Monovalent human live attenuated vaccine, 2-dose schedule

Explanation: Rotarix™ is a monovalent live attenuated human rotavirus vaccine given orally in a 2-dose schedule (at 2 and 4 months).

THANK YOU